

Davi GUANABARA DE ARAGÃO



PROFILE

Work focused on intelligent systems with emphasis on decision problems and machine learning. Experience in reinforcement learning, simulation-driven engineering, and applied research for autonomous systems.

CONTACT

@davi_guanabara@live.com
☎ +55 85 9 8673-6026
🌐 www.linkedin.com/in/davi-guanabara-91b3ab82/
🐙 github.com/DaviGuanabara
🌐 lattes.cnpq.br/5079017129840047

PERSONAL INFORMATION

Citizenship: **Brazilian**
Address: Sao Paulo, Brazil
Languages: **English (CEFR B2, DET score: 115)**

SKILLS

- Python (PyTorch)
- Reinforcement and supervised learning in simulation-driven settings
- Deep learning
- Modeling and experimentation in simulated environments

PROJECTS

SKYWEAVER

◊ Dynamic airspace configuration platform for UTM/U-Space. ◊ Primary research platform for dynamic airspace configuration. ◊ The public portfolio currently exposes the verified project scope and keeps the structure ready for future public architecture material.

REED

◊ Reactive runtime for deterministic dataflow execution. ◊ A runtime-oriented project centered on deterministic reactive computation. ◊ The structure is ready for deeper technical documentation as public repository materials become available.

ONDEVALE

◊ Web application that estimates the price per square meter of real estate in Sao Paulo using only geographic location. ◊ Developed as a data science case study for PO-235 at ITA and UNIFESP, with emphasis on methodology, validation, and deployment. ◊ The public repository separates dataset handling, KNN training artifacts, a FastAPI backend, and a Vue frontend for interactive map-based inference.

DEEP REINFORCEMENT LEARNING WITH GRAPH ATTENTION FOR COOPERATIVE LOYAL WINGMEN

◊ Published research on cooperative loyal wingmen under shared spatio-temporal observations. ◊ Research focused on multi-agent autonomy with graph attention mechanisms and richer situational awareness. ◊ The publication combines supervised pre-training with reinforcement-learning fine-tuning and uses graph attention networks to process shared observations.

DEEP LEARNING FOR GEO-TEMPORAL ENVIRONMENTAL MODELING

◊ Spatio-temporal data analysis and simulation framework using deep attention mechanisms and uncertainty-aware supervised learning workflows. ◊ Framework for simulating, analyzing, and predicting spatiotemporal phenomena from sparse sensor data. ◊ The framework centers on SensorManager, a Gymnasium-style environment, a wrapper-inspired training interface, and deep learning models built around sets of random neighbors and attention.

EDUCATION

PH.D. IN ELECTRONIC ENGINEERING AND COMPUTING (EEC) - INFORMATICS

Aeronautics Institute of Technology (ITA) **2025 - Present** ◊ Current doctoral research in data science, intelligent systems, and engineering.

GRADUATE CERTIFICATE IN DATA SCIENCE (CEDS)

Aeronautics Institute of Technology (ITA) **2024 - 2026** ◊ Graduate coursework in data science.

MASTER'S IN AERONAUTICAL COMPUTING (MPCOMP)

Aeronautics Institute of Technology (ITA) **2021 - 2024** ◊ Research-oriented work in deep reinforcement learning.

GRADUATE CERTIFICATE IN BUSINESS MANAGEMENT

Fundacao Getulio Vargas (FGV) **2019 - 2021** ◊ One of Brazil's leading institutions in business, economics, and public administration.

BACHELOR'S DEGREE IN CONTROL AND AUTOMATION ENGINEERING

University of Fortaleza (UNIFOR) **2013 - 2019** ◊ Undergraduate foundation in control, automation, and embedded systems.

PUBLICATIONS

DEEP REINFORCEMENT LEARNING WITH GRAPH ATTENTION FOR COOPERATIVE